Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/27/2018
PI Name:	Alwood, Joshua Ph.D.		
Project Title:	2012 Presidential Early Career Award for Scientist and Engineers (PECASE)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	joshua.s.alwood@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1490
Organization Name:	NASA Ames Research Center		
PI Address 1:	MS 261		
PI Address 2:	Building N261, Rm 113		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Space Biology PECASE AWARD
Start Date:	04/01/2015	**End Date:**	09/30/2018
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			
	Astronauts are exposed to weightlessness and ionizing radiation during spaceflight. To combat the bone loss that results from weightlessness, astronauts spend a substantial amount of time exercising. We aim to study these environmental conditions using basic biological models in ground based work that lays the ground work for future work. After the first two years, this Presidential Early Career Award for Scientist and Engineers (PECASE) award consisted of four projects: Project 1, Title: Epigenetic Regulation During Simulated Spaceflight: MicroRNA Expression, Histone Deacetylase Status, and Methylation-Specific Sequencing. Description and Aim: This work complements the methylation work performed in the Space Biology New Investigator Grant ("Bone Loss During Simulated Weightlessness: The Role of		

Task Description:	Osteoclasts," ended April 2015), by allowing ‘omics approaches to cover larger swaths of the genome (e.g., whole genome sequencing and RNA sequencing). Additionally, this project will extend the type of epigenetic mechanisms to be studied beyond DNA methylation. Specifically, we plan to analyze microRNA expression and status of histone acetylation. Hypotheses: Removal of mechanical loading or, oppositely, elevations of mechanical loading induce pro-bone resorbing or pro-bone forming changes in bone cell gene expression, respectively, via epigenetic mechanisms. Project 2, Title: Skeletal Sex Hormone Receptors and Simulated Spaceflight. Description and Aim: This work complements an upcoming space mission (PI : Joseph Tash, KUMC) and will define the skeletal response of male or female mice to simulated long-duration spaceflight in the context of sex hormones and their receptor levels in each compartment of the femur (cancellous and cortical). The assays used may include RNA-Seq, protein quantification in bone and blood, and immunostaining of histological sections of the femur for sex hormone and receptor levels. Hypotheses: Mechanical disuse reduces the levels of circulating sex hormones and their receptor levels in bone. These changes contribute to skeletal deterioration during long-duration spaceflight. This project has been folded into Joe Tash / Lane Christensen’s NNX15AB48G and progress reported with their annual reporting. Project 3, Title: Investigating the effects of ionizing-radiation exposure on bone’s collagen mechanical properties. Description: I am co-mentoring an NASA Space Technology Fellow and Doctoral Candidate from University of California (UC) Berkeley, Megan Pendleton. (Megan’s fellowship covers her stipend, but does not cover animal procurement. Thus, I am covering the latter and some related expenses.) This is a collaboration with Tony Keaveny (UC Berkeley) and Jean Sibonga (Johnson Space Center). Project 4, Title: The role of Nuclear Cytoskeleton in the Osteocytic Response to Simulated Weightlessness. Description and Aim: This project aims to investigate the role of nuclear-cytoskeletal elements (LINC, Linker of Nucleoskeleton and Cytoskeleton complexes) in the skeletal response to simulated weightlessness. Description: This is a collaboration with Gunes Uzer, Boise State Univ and is very early stage. I hosted an intern from his lab during the summer of 2017 and we have submitted grants for review to Space Biology Appendix G. We plan to investigate gene expression or protein levels indicative of changes in this critical and emerging cytoskeletal complex. In the third year of the award period, progress has been made on each project listed in the Task Description (see above). Additionally, one brand new collaboration commenced. Specifically: Project 5, Title: Characterizing the effects of chronic 2G centrifugation on the rat skeletal system. Description: This is a collaboration with Charles Fuller (UC Davis). Our long-term plan is to assess artificial gravity (AG) as a possible solution to mitigate these changes. In this study, one of our aims is to determine the skeletal acclimation to chronic centrifugation. We hypothesize that a 2G hypergravity environment causes an anabolic response in growing male rats. Specifically, we predict chronic 2G to increase tissue mineral density, bone volume fraction of the cancellous tissue and to increase overall bone strength. Systemically, we predict that bone formation markers (i.e., osteocalcin) are elevated and resorption markers (i.e., tartrate resistant acid phosphatase) are decreased or unchanged from controls. Projects conducted under this award addressed multiple Research Recommendations for Animal Experiments and Gaps in Knowledge stated in the Space Biology Science Plan and the National Academy of Sciences Decadal Survey, Recapturing Space: Life and Physical Sciences for a New Era of Exploration. The knowledge generated by this research queries the molecular responses to the high-turnover bone loss that astronauts experience while simultaneously weightless and exercising. This unique environment has similarities to Earth-bound sedentary or bed-ridden individuals. Additionally, this work investigates fundamental molecular responses, specifically, non-coding microRNA and DNA methylation responses to simulated spaceflight, which may improve our fundamental understanding of skeletal regulation and the organism’s response to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project addresses multiple Research Recommendations for Animal Experiments and Gaps in Knowledge stated in the Space Biology Science Plan and the National Academy of Sciences Decadal Survey, Recapturing Space: Life and Physical Sciences for a New Era of Exploration. The knowledge generated by this research queries the molecular responses to the high-turnover bone loss that astronauts experience with simultaneous weightlessness and exercising. This unique environment has similarities to Earth-bound sedentary or bed-ridden individuals. Additionally, this work investigates fundamental molecular mechanisms, specifically, non-coding RNA and DNA methylation responses to simulated spaceflight, which may improve our fundamental understanding of skeletal regulation and the organism’s response to spaceflight.
Task Progress:	In the third year of the award period, progress has been made on each project as described in the above Task Description section. Additionally, one brand new collaboration commenced that aims to characterize the effects of chronic 2G centrifugation on the rat skeletal system.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Johnson AK, Scott RT, Ronca AE, Hoban-Higgins TM, Fuller CA, Alwood JS. "Characterizing the effects of chronic 2G centrifugation on the rat skeletal system." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Touchstone H, Uzer G, Alwood JS. "The Role of Nuclear Cytoskeleton in the Osteocytic Response to Simulated Weightlessness." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017

Abstracts for Journals and Proceedings	Pendleton M, Sadoughi S, Li A, Liu J, O'Connell G, Alwood J, Keaveny T. "Effect of Spaceflight-relevant Ionizing Radiation on Mechanical Properties of Mouse Vertebra for Repetitive Loading." ASBMR 2017 (American Society for Bone and Mineral Research) Annual Meeting, Denver, CO, September 8-11, 2017. J Bone Miner Res 2017;32 (Suppl 1). , Sep-2017
Abstracts for Journals and Proceedings	Scott RT, Alwood J, Nalavadi M, Anand S, Shirazi-Fard Y, Castillo A. "Zoledronate and Mechanical Loading Treatments During Simulated Weightlessness: Cancellous Structure and Osteocellular Responses." ASBMR 2017 (American Society for Bone and Mineral Research) Annual Meeting, Denver, CO, September 8-11, 2017. J Bone Miner Res 2017;32 (Suppl 1). , Sep-2017
Abstracts for Journals and Proceedings	Emerzian SR, Pendleton M, Li A, Liu J, Tang SY, Alwood J, O'Connell G, Keaveny T. "Effect Of Ex Vivo Ionizing Radiation On Static And Fatigue Properties Of Mouse Vertebral Bodies." 2018 Orthopedic Research Society, New Orleans, Louisiana, March 10-13, 2018. 2018 Orthopedic Research Society, New Orleans, Louisiana, March 10-13, 2018. , Mar-2018
Awards	Scott RT, Alwood J, Nalavadi M, Anand S, Shirazi-Fard Y, Castillo A. "Young Investigator Award (to Ryan T. Scott), American Society for Bone and Mineral Research. ASBMR 2017 (American Society for Bone and Mineral Research) Annual Meeting, Denver, CO, September 8-11, 2017." Sep-2017